SUBHAJIT HALDER

IoT & Embedded Systems Developer | AI/ML & Raspberry Pi Enthusiast
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Portfolio Website

IoT & Embedded Systems Developer with hands-on experience in **Raspberry Pi, ESP32, sensor interfacing, and real-time embedded applications**. Built **AI-assisted IoT systems** involving **physiological sensors, GPIO control, and hardware-software integration**. Experienced in **Python, C/C++, embedded debugging, communication protocols (I2C/SPI/UART)**, and rapid prototyping for **intelligent systems** and **automation projects.**

SKILLS

- **Programming:** C, C++, Python, NASM, Java, Bash/Shell Scripting
- **Core Areas:** Embedded Systems, Internet of Things (IoT), Raspberry Pi, Arduino, ESP32/ESP-IDF, ARM Microcontrollers, Sensor Interfacing, Circuit Design, PCB Design & Soldering
- **Concepts:** Embedded Debugging, Data Structures & Algorithms, Operating Systems, Computer Networks, System Design, Machine Learning, Multithreading, Object-Oriented Programming
- **Tools & Technologies:** Git, Linux, CMake, Visual Studio, Arduino IDE, KiCAD, Proteus, OpenCV, TensorFlow, PyTorch, scikit-learn, MySQL, PostgreSQL, SQLite, Cisco Packet Tracer, VBA
- **Hardware and Protocols:** GPIO, I2C, SPI, UART, Sensor Calibration, Embedded Debugging, ADC/DAC

PROJECTS

IoT Music Mood Analysis and Therapy System
- Built Raspberry Pi based IoT system integrating MAX30102, GSR sensor (ADS1115), and MPU6050 for physiological data acquisition
- Implemented real-time sensor interfacing using I2C communication and Python-based data processing pipeline
- Designed modular embedded architecture supporting live sensor mode and simulation/testing mode
- Developed ML-assisted mood classification and automated audio therapy triggering system
- Performed sensor calibration, debugging, and hardware integration for stable real-time operation

GitHub: *IoT Music Mood Analysis and Therapy System Repository*

Sphoorti – Real-Time Speech to Braille Assistive System
- Led a 4-member team to develop a real-time Speech-to-Braille converter for visually impaired users using ESP32 and embedded tactile output mechanisms
- Contributed to hardware development including GPIO-controlled actuator interfacing and solenoid driver circuitry for Braille output
- Achieved low-latency real-time output and won multiple 1st prizes in technical competitions and innovation expos

GitHub: *Sphoorti Repository*

Operating Environment (OE) - Custom Shell & System Utilities
- Engineered a cross-platform shell in C/C++ with 15+ commands using a custom abstraction layer for Windows and Linux with modular architecture
- Developed system utilities including a text editor, Mathematical expression evaluator (Shunting Yard Algorithm), and custom file management commands

GitHub: *Operating Environment Repository*

Network Routing Simulation (DVR & LSR)
- Developed a network routing simulator implementing Distance Vector Routing (Bellman-Ford) and Link State Routing (Dijkstra) algorithms using multi-threaded Python socket programming
- Simulated dynamic routing behavior across multiple virtual nodes including route convergence, path optimization, and topology updates in distributed network environments

GitHub: *DVR LSR Implementation Repository*

WORK EXPERIENCE

Freelancer - Code Analyst & Developer | January 2025 - Present | Remote
- Delivered 35+ embedded systems, IoT automation, and software solutions for international clients across USA, Germany, Saudi Arabia, and Australia
- Developed systems including databases, network simulators, and automation tools with typical turnaround of 2-5 days per project
- Managed client requirements, project delivery, and end-to-end delivery for 3-5 concurrent projects

DeepTechSolutions - Consultant, Mentor & Developer | March 2025 - Present | Remote
- Developed 10+ network simulation and automation solutions using Cisco Packet Tracer, Python, and VBA
- Built 5+ advanced Excel automation tools and ML-based scripts for practical use cases
- Mentored 15+ students in C++, Python, and networking concepts

CERTIFICATIONS

- Computer Forensics (CFCT+) (Udemy, 2024)
- Advanced C/C++ Programming (Udemy, 2024)
- Cybersecurity (Udemy, 2024)
- Python Advanced (GUVI, 2024)

EDUCATION

- ***B.Tech – Information Technology*** | *Jalpaiguri Government Engineering College* | *2024-2028* |
  Current SGPA: *8.29 (Sem 3)*